Predicting Mortgage-Backed Securities Prepayment Using Machine Learning

UE24CS352A Machine Learning - Mini-Project | Team: Bhoomika Kshatriya (PES1UG24AM068) | Gaurav Girish (PES1UG24AM103) | Section: B | Problem #: 55

1. Problem Statement

A residential mortgage-backed security (MBS) pays investors the interest and principal collected from a pool of home loans. **Prepayment** - a borrower repaying part or all of a loan earlier than scheduled - stops future interest and cuts the investor's return, so predicting it matters. Industry usually models prepayment at *pool* level with hand-specified statistical models. We instead frame it as **loan-level binary classification**: given a loan's origination attributes, its current state and the macro-economic environment in a given month, will it be prepaid *that month* (1) or not (0)? We implement and compare logistic regression, SVMs, Gaussian Discriminant Analysis and a feed-forward neural network, following the reference study by Guo, Boccio and Cameron.

2. Dataset

Source: Freddie Mac Single-Family Loan-Level Dataset (~45.5M fixed-rate 30-year mortgages originated 1999-2020). *Origination* files hold static fields (credit score, DTI, LTV/CLTV, original UPB and rate, occupancy, purpose, channel); *monthly performance* files hold current UPB, loan age, months to maturity, current rate and the zero-balance code. The two are merged on loan id; macro features (mortgage rate, house-price change, unemployment) are joined by month; **label = 1** when the zero-balance code is 01 (voluntary payoff); because code 01 also covers maturity, only 30-year (360-month) loans are kept. Missing sentinels (999/9999) become NaN and are median-imputed; categoricals are one-hot encoded; numeric features are standardised inside each model pipeline.

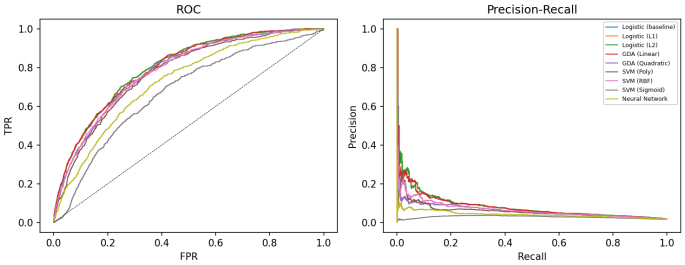
NOTE - synthetic data: the numbers below come from the bundled synthetic loan-month generator (same schema as Freddie Mac, prepayment hazard driven by rate incentive, balance, age, credit score, LTV, DTI). They validate the pipeline and are not Freddie Mac results. **Run reported here:** 200,911 loan-month rows from 3,000 loans, 23 features after one-hot encoding, prepayment rate 1.47% (imbalance 1:67). Split 60/20/20 by *loan* (train 121,275 / val 43,916 / test 35,720; 593 positive test rows).

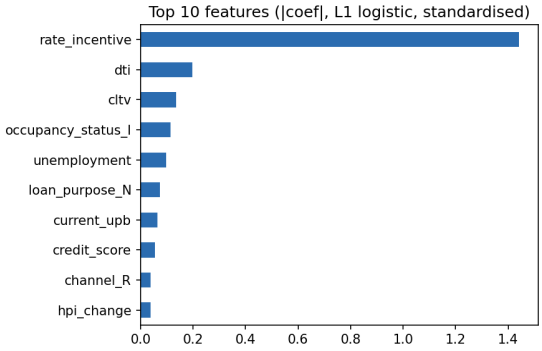
3. Approach and Implementation

- **Logistic regression** - baseline, L1 (feature selection) and L2 (multicollinearity) regularised.
- **SVM** - polynomial, RBF and sigmoid kernels; trained on a subset because the kernel matrix grows quadratically.
- **GDA** - linear (shared covariance, i.e. LDA) and quadratic (class-specific covariance, QDA).
- **Neural network** - feed-forward MLP (128-256-128, ReLU, Adam, early stopping), adapted from the paper's 3-layer net; positives oversampled 10x in place of the paper's positive loss weight of 10.
- **Evaluation under imbalance** - accuracy is misleading at 1:67 (predicting 'never prepay' scores 98.5%), so we also report precision, recall, F1, ROC-AUC and PR-AUC. Each model's decision threshold is tuned on the validation split (max F1) and applied once to the test split.
- **Code layout** - `src/data.py` (loader, synthetic generator, loan-grouped split), `src/models.py` (model zoo), `src/train.py` (train, evaluate, plots), `src/demo.py` (live scoring), `src/macro.py` (macro features from FRED), `src/tune.py` (hyper-parameter sweep), `tests/` (pytest suite for the data pipeline), `notebooks/eda.ipynb` (exploratory analysis), `report/` (this PDF and the deck).

4. Results (test split)

Model	Acc.	Prec.	Recall	F1	ROC-AUC	PR-AUC
Logistic (baseline)	0.9406	0.093	0.293	0.141	0.801	0.084
Logistic (L1)	0.9380	0.092	0.307	0.141	0.801	0.084
Logistic (L2)	0.9405	0.093	0.293	0.141	0.801	0.084
GDA (Linear)	0.9402	0.093	0.297	0.141	0.797	0.082
GDA (Quadratic)	0.9526	0.094	0.214	0.130	0.784	0.063
SVM (Poly)	0.8751	0.056	0.410	0.098	0.772	0.058
SVM (RBF)	0.9413	0.081	0.245	0.121	0.783	0.066
SVM (Sigmoid)	0.8013	0.034	0.406	0.064	0.673	0.028
Neural Network	0.9467	0.063	0.160	0.091	0.725	0.042





5. Conclusions

- Best ranking quality (PR-AUC): **Logistic (L1)** (0.084); best F1: **GDA (Linear)** (0.141). Regularisation barely changed logistic performance, but L1 shrinks the feature set.
- The most influential feature in the L1 model is **rate incentive**, consistent with finance theory (borrowers refinance when the market rate falls below their loan rate).
- Under heavy imbalance, accuracy alone hides poor prepayment recall; PR-AUC / F1 and a tuned threshold give an honest comparison.
- **Tuning** (chosen on validation, scored once on test): QDA needs strong covariance shrinkage because some features are exact duplicates (reg_param 0.1 to 0.6 lifts its test PR-AUC from 0.063 to 0.078); logistic regression is insensitive to C, so the model ranking does not change.
- **Leakage control:** Freddie Mac reports current UPB = 0 in the payoff month, which reveals the label (the reference paper's top feature was current UPB). Our loader therefore uses the balance at the *start* of the month (previous month's UPB).
- **Limitations / future work:** kernel SVMs see only a subset of the data; next steps are time-based validation, an ensemble of the models, and comparison with pool-level models.

Reference: C. Guo, J. Boccio, C. Cameron, 'Predicting Mortgage Backed Securities Prepayment Using Machine Learning Methods', CS229 project report, Stanford University, Spring 2021 (course project, not peer-reviewed), https://cs229.stanford.edu/proj2021spr/report2/81995903.pdf. Code: https://github.com/gaurav-2030/mbs-prepayment.